This document is exploratory and is not yet linked from the main project index (README.md or GUIDE.md). It covers the ZP-J valuation bridge work (ZPJ_Scale, ZPJ_SelfApp, ZPJ_AczelConn, ZPJ_Model) not covered in the main ZP-J companion. For the main project, see README.md.

ZP-J Valuation Bridge — Exploratory Companion

Why ⊥ Is the Only Fixed Point

Deriving Self-Reference from Arithmetic

ZP Companion | Version 1.0 | May 2026

This companion explains the ideas in plain language. It covers the ZP-J valuation bridge — four Lean sub-files that derive AFA self-containment from valuation arithmetic. It is exploratory and not a formal document.

What This Document Does

ZP-J proved T-EXEC: the bottom element ⊥ of any ZP-A semilattice with AFA grounding is the unique Quine atom ($\bot = \{\bot\}$). But T-EXEC imported three facts from AFA as class fields — axioms in the Lean encoding: selfMem (self-membership as a proposition), bot_self_mem (⊥ is self-containing), and quine_unique (any two self-containing elements are equal). These appear as axioms because the abstract lattice lacks the machinery to derive them from anything more primitive.

The four ZP-J sub-files ask the next question: can those AFA class fields be derived from something more basic? The answer is yes — from a valuation structure. If a scale operation strictly increases a measure, then ⊥ (the element of infinite measure) is forced to be the unique fixed point of scale. The specific AFA content needed here — self-membership, ⊥ as the unique self-containing element, and uniqueness — follows as a theorem, not an assumption.

The Valuation Argument

A valuation assigns a numeric measure to each element of the lattice. The bottom element ⊥ has measure ∞ — represented as ⊤ (the top element of ℕ∞, the extended naturals). Every other element has a finite measure. Elements with higher measure are structurally closer to ⊥ in the valuation; ⊥ itself sits at measure ∞. Formally: $\mathrm{val}(\bot) = \top$, and $\mathrm{val}(x) \neq \top$ for $x \neq \bot$.

The scale operation moves each element one step further in the valuation — strictly increasing the measure. Formally: $\mathrm{val}(\mathrm{scale}\ x) = \mathrm{val}\ x + 1$ for every $x \neq \bot$. This single axiom has one powerful consequence: scale cannot have a fixed point anywhere other than ⊥.

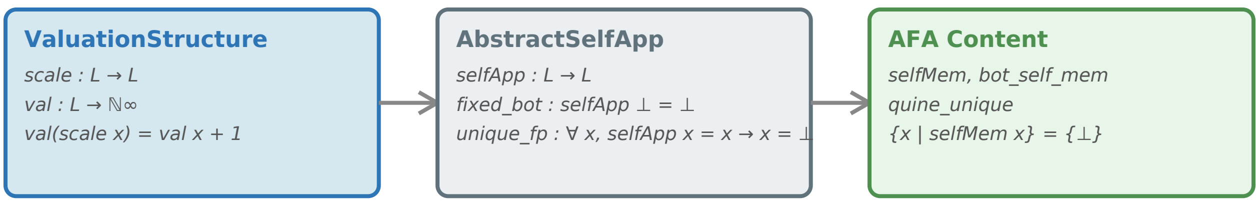
The proof is a one-line contradiction. Suppose $\mathrm{scale}\ x = x$ and $x \neq \bot$. Then $\mathrm{val}(x) = \mathrm{val}(\mathrm{scale}\ x) = \mathrm{val}(x) + 1$. But val(x) is finite (since $x \neq \bot$), and no finite number equals itself plus one. Contradiction. So the only fixed point of scale is ⊥. (The exception "$x \neq \bot$" is intentional: at ⊥ itself, scale fixes ⊥ by axiom scale_bot, and $\infty + 1 = \infty$ in ℕ∞ is consistent — the +1 rule characterises non-bot elements, not ⊥ itself.)

The impossible equation

In the natural numbers $\mathbb{N}$: if $n = n + 1$, what is n? There is no solution. Adding 1 always moves you forward — you can never stay in place. Only at the extended point $\top$ in $\mathbb{N}\infty$ does this equation have a solution: $\top + 1 = \top$ in $\mathbb{N}\infty$ (successor saturates at $\top$ — WithTop arithmetic by convention). The valuation argument is the same: val(scale x) = val x + 1 means scale always moves forward in the measure. The only escape is already being at measure ∞ — which means $x = \bot$.

The Derivation Chain

The valuation argument runs through three layers. Each step takes the output of the previous one and derives something stronger:



Sorry-free derivation chain. Each arrow replaces an AFA axiom with a theorem proved from valuation arithmetic alone.

Three-step sorry-free derivation. ValuationStructure provides the arithmetic input. AbstractSelfApp extracts the minimal fixed-point structure. AFA Content is the output: self-membership, $\bot$ as the unique self-containing element.

Step 1 — ValuationStructure → AbstractSelfApp. ZPJ_Scale.lean builds the instance toAbstractSelfApp: set selfApp = scale, then derive fixed_bot from scale_bot and unique_fp from the one-line valuation contradiction above. AbstractSelfApp has just two fields: a self-application operation and the claim that $\bot$ is its unique fixed point.

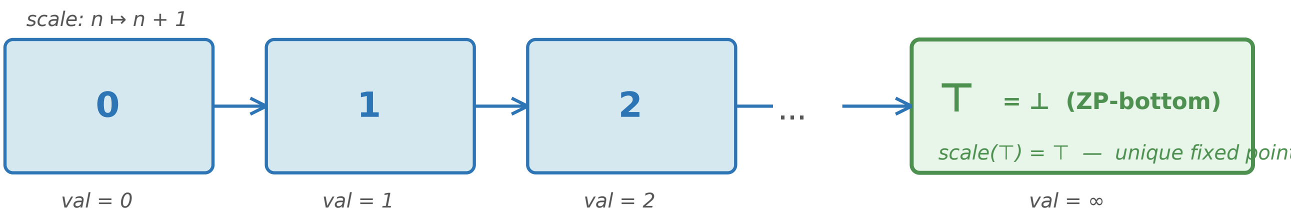
Step 2 — AbstractSelfApp → AFA Content. ZPJ_SelfApp.lean derives from fixed_bot and unique_fp alone: selfMem x := (selfApp x = x); bot_self_mem ($\bot$ is self-containing, from fixed_bot); quine_unique (any two self-containing elements are equal, from unique_fp); and $\{x \mid \text{selfMem } x\} = \{\bot\}$ — the self-containing set has exactly one element.

All steps are sorry-free. Every result depends only on [propext, Classical.choice, Quot.sound] — no AFA axioms are imported. AFA content is derived from the arithmetic of val and scale.

The Concrete Model: ℕ∞

The derivation chain works for any type carrying a ValuationStructure. ZPJ_Model.lean provides the concrete witness: $\mathbb{N}\infty$ (the extended natural numbers, WithTop $\mathbb{N}$ in Lean 4) with a specific choice of join, bottom, scale, and valuation.

ZPSemilattice uses a reversed order on ℕ∞. In the ZP partial order, $x \leq_{ZP} y$ iff min x y = y iff x ≥ y in ℕ∞'s natural order. Under this reversed order, min becomes the join (least upper bound in the ZP lattice), and ⊤ — the natural maximum of ℕ∞ — becomes the ZP-bottom. So: join = min, bot = ⊤, val = identity function (val n = n, val ⊤ = ⊤), scale = successor (n → n + 1).



ℕ∞ with join = min and bot = ⊤: the ZP order reverses ℕ natural order. ⊤ is the unique fixed point of scale — ZP-bottom (⊥).

ℕ∞ with scale = successor. Each n ∈ ℕ has val n = n; scale moves n to n+1. ⊤ is the unique fixed point: ⊤ + 1 = ⊤ in ℕ∞ (WithTop arithmetic). ⊤ is ZP-bottom (⊥).

The theorem natInf_selfMem_singleton states: {x : ℕ∞ | x + 1 = x} = {⊤}. This is AFA content derived entirely from arithmetic: the "self-containing set" in this model is exactly {⊤} = {⊥}. The proof delegates to val_selfMem_singleton from ZPJ_Scale.lean — the same abstract result that applies to any ValuationStructure.

> The ZP order on ℕ∞ is reversed: ⊤ is the bottom (most open, val = ∞), and 0 is the maximum (fully constrained, val = 0). Scale moves elements toward ⊤ — toward less constraint, toward the bottom. This is the "indexed family" picture: each level n is a state of constraint, and ⊤ is the unreachable limit where all valuation chains converge.

The 2-Adic Parallel

The valuation structure was motivated by the 2-adic integers $\mathbb{Z}_2$. In $\mathbb{Z}_2$: scale x = 2 × x, and val x = $v_2(x)$ — the 2-adic valuation, the exponent of the highest power of 2 dividing x. The 2-adic valuation assigns infinite measure to 0 by convention: $v_2(0) = \infty$. This is consistent with the fact that 2^n divides 0 for every n, but the infinitary value is part of the definition of the extended 2-adic valuation, not derived from divisibility alone.

The key identity: $v_2(2x) = v_2(x) + 1$ for $x \neq 0$. Scale (×2) strictly increases the 2-adic valuation by exactly 1 — the same axiom as val_scale in ValuationStructure, now in $\mathbb{Z}_2$. The unique fixed point of ×2 is 0, proved by ring arithmetic: 2x = x gives x = 2x − x = 0 in $\mathbb{Z}_2$.

$\mathbb{Z}_2$ is not a ZPSemilattice (it is a ring, not a join-semilattice), so it cannot be a formal ValuationStructure instance. ZPJ_Scale.lean §V proves all four ValuationStructure axioms for $\mathbb{Z}_2$ as standalone theorems — sorry-free, using Mathlib's PadicInt library. The ℕ∞ model provides the formal instance; $\mathbb{Z}_2$ shows the same abstract structure in a mathematically rich concrete object.

Why $\mathbb{Z}_2$ and not ℝ?

In $\mathbb{R}$, the only compatible valuation is the trivial one ($v(x) = 0$ for all $x \neq 0$, $v(0) = \infty$), which gives no useful scale operation: $v(2x) = v(x)$ for all $x \neq 0$. There is no canonical non-Archimedean valuation on $\mathbb{R}$ satisfying $v(2x) = v(x) + 1$. $\mathbb{Z}_2$ carries exactly such a valuation — one where 0 has infinite measure and multiplication by 2 shifts every other element one step up. This is why the fixed-point argument works in $\mathbb{Z}_2$ and not in $\mathbb{R}$.

DC-Free: Identification Replaces Construction

Aczel's original proof that the largest fixed point of a set-continuous operator Φ is $J_\Phi = \cup\{x \mid x \subseteq \Phi(x)\}$ used the Axiom of Dependent Choice (DC) to construct an ω-chain (Non-Well-Founded Sets, 1988, ch. 6, p. 77). Aczel noted explicitly: "I do not know if this use of the axiom of dependent choices was essential."

ZPJ_AczelConn.lean answers this for the self-membership case: DC is not needed. quine_unique — AFA's uniqueness clause — directly forces any self-containing element to equal ⊥. If x is self-containing and ⊥ is self-containing, then x = ⊥ in one step. No chain needed. Identification replaces construction.

The valuation bridge has the same structure. unique_fp says: if scale x = x then x = ⊥. The fixed point is identified by the one-line valuation contradiction — not by constructing a sequence converging to it. DC-free. This matters because DC is a non-trivial existence principle. DC-free proofs work in settings where choice is restricted, and formally they carry a lighter axiom footprint.

Scope: ZPJ_AczelConn.lean proves DC is unnecessary for the self-membership case specifically. Whether it can be eliminated for all set-continuous operators Φ remains open. One natural conjecture — not established in the literature — is that DC-eliminability depends on whether uniqueness holds for the specific Φ. ZP provides a positive answer for the case ZP cares about.

Key Results — ZP-J Valuation Bridge

ValuationStructure (ZPJ_Scale.lean): val(scale x) = val x + 1 forces ⊥ to be the unique fixed point of scale — AFA content derived, no AFA axioms imported. ✓ AbstractSelfApp (ZPJ_SelfApp.lean): minimal fixed-point typeclass from which selfMem, bot_self_mem, quine_unique, {x | selfMem x} = {⊥} are theorems. ✓ ℕ∞ model (ZPJ_Model.lean): concrete ValuationStructure — natInf_selfMem_singleton : {x : ℕ∞ | x+1=x} = {⊤}. ✓ 2-adic parallel (ZPJ_Scale.lean §V): q2Val_scale proved sorry-free from Mathlib PadicInt. ✓ DC-free (ZPJ_AczelConn.lean): identification replaces construction for self-membership. ✓ All results: [propext, Classical.choice, Quot.sound] — no sorryAx anywhere.